DIPLOMA PROGRAMME PRACTICE · SL · SET A

Biology / Paper 1B

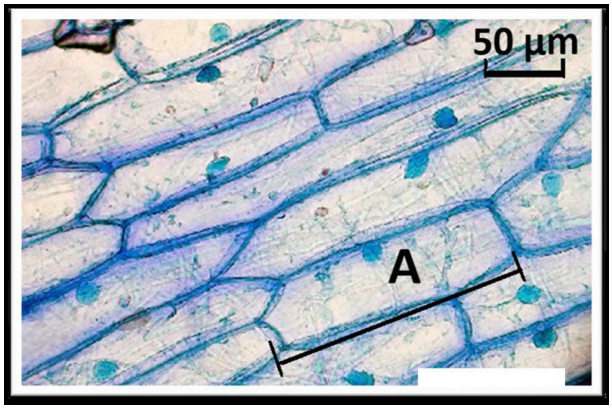
Question booklet · 25 marks · 90 minutes across 1A and 1B

Show working, units and suitable precision. Use a calculator. Use separate answer paper for additional working and diagrams.

QUESTION 1 [6]

Measuring a stained epidermis

The light micrograph shows epidermal cells. The black bracket across cell A and the scale bar belong to the same image. Work from the two lengths in the figure. On a separate display the scale bar measures 12.0 mm.



Stained epidermal tissue, showing bracket A and a 50 µm scale bar. Supplied SL Paper 1B model answers, p. 2.

(a) Estimate the bracketed length of cell A. Show the ratio you used. **[2]**

(b) Calculate the magnification on the separate display. **[2]**

(c) Give one visible feature supporting a plant origin. Explain why the dark boundary alone does not distinguish the plasma membrane from the cell wall at this resolution. **[2]**

QUESTION 2 [6]

Ventilation and exchange

In a simplified lung model, the first 0.15 dm³ of each breath supplies conducting airways and does not reach exchange surfaces. All volumes are measured under the same conditions.

Condition	Breaths min⁻¹	Volume per breath / dm³
Rest	12	0.50
Steady exercise	24	1.10

(a) Calculate the volume reaching exchange surfaces each minute at rest and during exercise. **[2]**

(b) Under the model, compare slow breathing at 10 breaths of 0.60 dm³ per minute with the resting pattern. Explain the difference despite equal total ventilation. **[2]**

(c) Explain why more ventilation need not produce a proportional increase in oxygen uptake by the body. **[2]**

QUESTION 3 [6]

Separating enzyme activity from background change

A pigment forms as an enzyme converts a substrate. Absorbance is proportional to pigment concentration in this range. The table gives changes in absorbance over the first 40 s. Matched tubes without enzyme measure background pigment formation.

Temperature / °C	With enzyme	Without enzyme
15	0.100	0.020
25	0.190	0.030
35	0.260	0.060
45	0.245	0.105

(a) Calculate the corrected mean absorbance change per second at 35 °C. **[2]**

(b) Compare the corrected rates at 25 °C and 45 °C with evidence. **[2]**

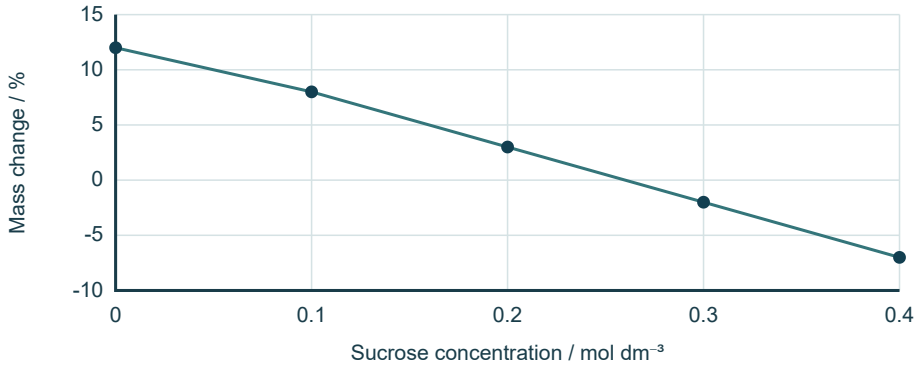
(c) Evaluate the claim that 35 °C is the enzyme’s exact optimum. **[2]**

QUESTION 4 [7]

An isotonic estimate from plant tissue

Cylinders cut from comparable regions of one potato were immersed for equal times. At each sucrose concentration the mean percentage mass change was measured. The graph joins means to guide interpolation.

Sucrose / mol dm⁻³	0.00	0.10	0.20	0.30	0.40
Mass change / %	+12	+8	+3	−2	−7



Mass change / % plotted against Sucrose concentration / mol dm⁻³. Points are measured means or observations as tabulated.

(a) Estimate the concentration for zero net mass change by linear interpolation. **[2]**

(b) A matched cylinder initially weighs 2.50 g. Predict its final mass at 0.40 mol dm⁻³ and explain the sign of the change. **[2]**

(c) Explain one reason to blot samples consistently and one limitation of using all cylinders from a single tuber. **[2]**

(d) State what continues at the membrane at the zero crossing. **[1]**